

Strategic alignment: EA defines how AI agents support business capabilities and strategic goals. This includes identifying high-impact use cases, prioritising agent development, and measuring business value.

Architectural governance: EA establishes standards and guidelines for agent development, deployment, and management. This ensures the agent ecosystem's consistency, security, scalability, and compliance.

Integration and interoperability: EA designs the agent architecture to ensure seamless integration with existing enterprise systems, data sources, and applications. This involves defining APIs, data flows, and communication protocols.

Technology selection and standardisation: EA evaluates and recommends open source frameworks, tools, and platforms for building AI agents, promoting standardisation, and reducing technology sprawl.

Risk management and security: EA addresses security, privacy, ethical, and legal considerations related to AI agents. This includes defining security protocols, data governance policies, and ethical guidelines.

Enterprise architecture is the central orchestrator for AI agent initiatives within an enterprise. It ensures that AI agents are deployed effectively, securely, and in a manner that genuinely benefits the organisation's strategic objectives. This is essential for realising the full potential of AI agents in the enterprise landscape.

Top open source offerings for building generative AI agents

The open source ecosystem is brimming with powerful tools and frameworks that empower enterprises to build their own generative AI agents. Navigating this landscape requires understanding the different categories of tools and identifying the best fit for specific needs. The following list categorises and highlights some

of the most prominent open source offerings, providing a starting point for enterprises exploring this space.

Agent frameworks (orchestration and management)

Agent frameworks are essential for structuring and managing the complexities of AI agents. These frameworks provide the scaffolding for building intelligent workflows, handling agent interactions, and managing the overall agent lifecycle. Here are some leading open source agent frameworks.

LangChain: A widely adopted framework designed to simplify the development of LLM-powered applications, it provides modules for model integration, prompt management, memory, agent creation, and chains of operations. LangChain excels in building complex agent workflows through its flexible chaining mechanisms and extensive integrations.

AutoGen: Microsoft focuses on enabling the development of AI agents that can collaborate with humans to solve complex tasks. AutoGen stands out for its multi-agent conversation capabilities, which allow for sophisticated task decomposition and collaborative problem-solving.

Haystack (Deepset): An open source framework for building search systems and conversational AI applications. Haystack provides robust components for document retrieval, question answering, and agentic interactions, making it suitable for knowledge-intensive agent applications.

These agent frameworks offer distinct strengths and cater to different agent development needs. LangChain provides broad flexibility and a mature ecosystem, AutoGen excels in collaborative multi-agent scenarios, and Haystack is particularly well-suited for knowledge-driven applications. Choosing the proper framework depends on the enterprise's

requirements and the type of AI agents being developed.

LLM model libraries and APIs

At the heart of any generative AI agent lies the large language model (LLM). Open source offers diverse LLMs, each with their own strengths and characteristics. Accessing and leveraging these models is facilitated by model libraries and APIs. The following are key open source resources for LLMs.

Hugging Face Transformers: This is a central hub for pre-trained models, datasets, and tools for natural language processing. It offers access to thousands of open source LLMs, including models like Llama 2, Falcon, and more, providing a vast selection for agent development.

vLLM: An open source library for fast LLM inference and deployment, vLLM optimises performance and efficiency, making it ideal for deploying agents requiring real-time responsiveness and scalability.

OpenAI compatible APIs (e.g., Together AI, Anyscale endpoints): While OpenAI itself is not open source, several platforms offer open source models accessible through OpenAI-compatible APIs. These balance ease of use with access to powerful open source LLMs.

These open source LLM resources provide enterprises with a broad spectrum of choices. Hugging Face Transformers offers an extensive catalogue of models, vLLM prioritises performance, and OpenAI-compatible APIs provide accessibility and ease of integration. The selection of the LLM resource should be guided by the agent's specific performance requirements, desired model characteristics, and integration preferences.

Tools for development and deployment

Building and deploying generative AI agents for enterprise use requires robust development and deployment infrastructure. Open source tools provide